

Jaymin Nirmal

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 3 years of experience in predictive modeling, NLP, claims analysis, and advanced analytics across healthcare, manufacturing, banking, and retail. Skilled in Python, SQL, Spark, Databricks, Azure, Snowflake, Tableau, and Power BI, with expertise in logistic regression, XGBoost, Random Forest, anomaly detection, clustering, and time-series forecasting. Proven track record of reducing hospital readmissions, detecting fraud, improving equipment uptime, and boosting marketing ROI. Collaborative problem-solver with strong communication skills, experienced in turning complex data into actionable strategies for cross-functional teams.

TECHNICAL SKILLS

Predictive Modeling & Analytics: Logistic Regression, Random Forest, XGBoost, Clustering, Time-Series Forecasting, Anomaly Detection, Feature Engineering

Data Engineering & Processing: ETL Development, Data Pipelines, Azure Databricks, Snowflake, Azure Data Factory, Spark, SQL

NLP & Text Analytics: Named Entity Recognition, Text Classification, Transcript Analysis, spaCy, HuggingFace

Visualization & Reporting: Tableau, Power BI, Interactive Dashboards, Automated Reporting, KPI Tracking

Programming & Tools: Python (Pandas, NumPy, Scikit-learn), SQL, Spark SQL, R, Excel

Collaboration & Domain Expertise: Healthcare Claims Analysis, Patient Engagement, Fraud Detection, Customer Churn, Predictive Maintenance, Cross-Functional Team Support

PROFESSIONAL EXPERIENCE

United Healthcare Group, USA | Data Scientist

Jan 2025 – Present

- Built predictive risk models on large-scale claims, EHR, and social data using Python, SQL, Spark, and Databricks, helping care teams identify members at high risk of rehospitalization or disease progression.
- Improved model accuracy with logistic regression, XGBoost, and feature engineering, which enhanced risk stratification and supported early interventions that reduced readmissions.
- Designed scalable pipelines in Azure Databricks to process de-identified claims and clinical records, ensuring secure and efficient data handling for predictive analytics.
- Developed NLP workflows in Python with spaCy and HuggingFace to analyze physician notes, EHR summaries, and call transcripts, extracting medical and social insights.
- Applied named entity recognition and text classification to detect symptoms, care gaps, and social needs (e.g., food insecurity, transport issues), improving automated identification of members needing support by 30–50%.
- Partnered with clinical and service teams to turn predictive and NLP findings into actionable strategies, improving patient engagement and support programs.

Hexaware Technology, India | Data Scientist

Mar 2021 to July 2023

- Built predictive maintenance models in Python, Scikit-learn, Spark, and Databricks on IoT telemetry and maintenance logs, cutting downtime by 20–30% and repair costs by 25%.
- Developed ETL pipelines in Azure Data Factory and Snowflake to process high-frequency sensor and operational data, ensuring scalable and automated data preparation.
- Applied time-series feature engineering (rolling stats, anomaly scores) and trained Random Forest and XGBoost models, achieving ~85–90% accuracy in predicting failures.
- Created customer churn models with Python and SQL on transaction, demographic, and service usage data in Azure Synapse/Databricks, reducing attrition by 15%.
- Built fraud detection pipelines using anomaly detection (isolation forests) on large transaction datasets, identifying >80% of unauthorized activities.
- Delivered interactive dashboards in Tableau and Power BI to track predictive outputs, equipment health, ROI, and campaign performance for faster business decisions.
- Performed market-mix modeling and segmentation (multivariate regression, K-means clustering) on CRM, POS, and ad spend data, driving 15–20% ROI growth and a 10% sales uplift.
- Developed a personalization framework with collaborative filtering and NLP-based content analysis of product descriptions, enabling tailored offers and stronger customer engagement.

EDUCATION

Master of Science, Data Science | Stevens Institute of Technology, Hoboken, New Jersey

Bachelor of Technology, Computer Science and Engineering | Ganpat University, Ahmedabad, India

CERTIFICATES

- Big Data Integration and Processing
- AWS Academy Cloud Foundations – Amazon Web Services (AWS)
- Google Cloud Big Data and Machine Learning Fundamentals
- AWS Academy Graduate – Machine Learning Foundations